Circles and Parabolas

Section 1: Standard Equation and Features of Circles

1 Find the center and radius of the circle $(x-3)^2+(y+5)^2=49$.

a. Center: $(3,\ -5)$, Radius: 7

b. Center: $(-3, 5)$, Radius: 7

c. Center: $(3,\ -5)$, Radius: 49

d. Center: $(-3, 5)$, Radius: 49

2 Convert the general equation $x^2+y^2-6x+8y-11=0$ to standard form by completing the square.

a. $(x-3)^2+(y+4)^2=36$

b. $(x+3)^2+(y-4)^2=36$

c. $(x-3)^2+(y+4)^2=25$

d. $(x-6)^2+(y+8)^2=11$

3 Find the equation of the circle passing through the origin with center $(-2, 3)$.

a. $(x+2)^2+(y-3)^2=13$

b. $(x-2)^2+(y+3)^2=13$

c. $(x+2)^2+(y-3)^2=5$

d. $(x+2)^2+(y-3)^2=25$

Section 2: Parabolas with Vertical Axis of Symmetry

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4 Find the focus and directrix of the parabola $x^2 = 12y$.

a. Focus: $(0, 3)$, Directrix: $y = -3$

b. Focus: $(3, 0)$, Directrix: $x = -3$

c. Focus: $(0, 12)$, Directrix: $y = -12$

d. Focus: $(0, 6)$, Directrix: $y = -6$

5 Find the standard equation of the parabola with vertex $(0, 0)$ and focus $(0, -2)$.

a. $x^2 = -8y$

b. $y^2 = -8x$

c. $x^2 = 8y$

d. $y^2 = -4x$

6 Find the focal diameter (latus rectum length) of $(x - 1)^2 = -16(y + 2)$.

a. 16

b. 4

c. 8

d. 32

Section 3: Parabolas with Horizontal Axis of Symmetry

7 Find the vertex, focus, and directrix of $(y - 2)^2 = 8(x + 1)$.

a. Vertex: $(-1, 2)$, Focus: $(1, 2)$, Directrix: $x = -3$

b. Vertex: $(1, -2)$, Focus: $(3, -2)$, Directrix: $x = -1$

c. Vertex: $(-1, 2)$, Focus: $(-1, 4)$, Directrix: $y = 0$

d. Vertex: $(2, -1)$, Focus: $(4, -1)$, Directrix: $x = 0$

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8 Find the equation of a parabola with vertex $(3, -1)$ and focus $(1, -1)$.

a. $(y+1)^2 = -8(x-3)$

b. $(x-3)^2 = -8(y+1)$

c. $(y+1)^2 = 8(x-3)$

d. $(y-3)^2 = -4(x+1)$

9 Convert $y^2 + 4y - 4x + 8 = 0$ to standard form.

a. $(y+2)^2 = 4(x-1)$

b. $(y-2)^2 = 4(x+1)$

c. $(y+2)^2 = 4x - 8$

d. $(x-1)^2 = 4(y+2)$

Section 4: Reflective Properties and Applications

10 A satellite dish has the shape of a paraboloid. If the cross-section is $x^2 = 20y$ (in inches), where should the receiver be placed?

a. At $(0, 5)$, which is 5 inches above the vertex

b. 20 inches above the vertex

c. 10 inches above the vertex

d. 2.5 inches above the vertex

11 A suspension bridge cable hangs in the shape of a parabola with towers 100 m apart and 20 m high, touching the road at the center. Find its equation with origin at the vertex.

a. $y = \frac{1}{125}x^2$

b. $y = \frac{1}{250}x^2$

c. $y = \frac{1}{50}x^2$

d. $y = \frac{1}{20}x^2$

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12 Why do parabolic headlights use a light source placed exactly at the focus?

a. Light rays reflecting off the parabolic surface emerge in a parallel beam

b. It spreads the light in all directions equally

c. It minimizes heat production

d. It prevents the mirror from tarnishing

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